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Buyers' Sourcing Strategies and Suppliers' Markups in Bangladeshi Garments

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1 Big picture

- **Question.** Do suppliers earn different markups from buyers with different sourcing strategies, and what do those markup differences reveal about global value chain relationships?
- **Empirical setting.** The paper studies Bangladeshi woven garment exporters. Bangladesh is a leading garment exporter, and the industry has rich buyer-seller relationships with global retailers and brands.
- **Core distinction.** Buyers differ between **spot sourcing**, where orders are spread across many suppliers and allocated arm's length, and **relational sourcing**, where orders are concentrated among a small number of suppliers and supported by long-term relational incentives.
- **Main theoretical mechanism.** Relational buyers pay higher prices and markups to compensate suppliers for costly actions that ensure reliable delivery when shocks occur.
- **Main empirical challenge.** Testing this requires buyer-order-level prices and buyer-order-level costs. Prices are observable in customs data, but costs are usually not.
- **Data breakthrough.** The authors use novel order-level production data that match export orders to the fabric used and sewing-line labor employed, allowing them to study order-specific variable input use.
- **Identification insight.** The same supplier can produce the same product in the same year for different buyers. Seller-product-year fixed effects therefore compare orders that are extremely similar except for buyer identity and sourcing strategy.
- **Main finding 1.** Relational buyers pay higher prices than spot buyers for otherwise comparable orders.
- **Main finding 2.** Relational buyers do not systematically use more expensive fabric, more fabric, more labor, or more skilled workers.
- **Main finding 3.** Because marginal costs do not increase, the higher prices paid by relational buyers translate into higher supplier markups.
- **Quantitative interpretation.** A one standard deviation shift in the sourcing metric, comparable to moving from an average buyer to The Gap's relational sourcing, is associated with an economically meaningful increase in markups and relationship value.
- **Policy implication.** Relational buyers create positive pecuniary externalities for suppliers. A planner may want to attract buyers with relational sourcing capabilities because individual buyers do not internalize the broader gains from upgrading supplier relationships.

2 Incremental contribution of the paper

2.1 Contribution relative to global value chain research

- Prior work emphasizes that global buyers affect developing-country suppliers through learning, quality upgrading, standards, and access to export markets.
- This paper adds a direct markup channel: the way a buyer sources orders changes the rents earned by suppliers.
- It shows that buyers are not interchangeable demand sources. Their organizational capabilities and sourcing strategy affect the division of surplus in trade relationships.

Interpretation: The paper moves the literature from asking whether exporters benefit from global buyers to asking which buyers create supplier rents and why.

2.2 Contribution relative to empirical markup measurement

- Standard markup estimation usually works at firm, product, or plant level and relies on production-function assumptions using aggregate inputs.
- This paper measures markup differences across *orders* produced by the same seller for different buyers.
- The novelty is matching export prices to order-level variable input quantities and prices, especially fabric and sewing labor.

Interpretation: The contribution is methodological: the paper creates a way to compare markups across buyers within seller-product-time cells, rather than only across firms or products.

2.3 Contribution relative to relational contracting theory

- Relational contract theory predicts that repeated relationships can sustain noncontractible actions by promising future rents.
- The paper takes this prediction to data in a global supply chain and shows that relational buyers pay higher markups consistent with incentive provision.
- It also documents that sourcing style is partly a buyer-level organizational attribute, not merely a pair-specific feature.

Interpretation: The paper provides direct evidence that relational sourcing is priced in markups and that buyer capabilities matter for forming such relationships.

2.4 Contribution relative to trade and development policy

- Existing policy discussions often emphasize attracting foreign buyers or increasing export participation.
- This paper shows that the type of buyer matters. Relational buyers may generate supplier rents and support upgrading, while spot buyers may squeeze markups.
- Because relational capabilities are buyer-level attributes, policy can target buyer entry and relationship formation rather than trying to micromanage individual contracts.

Interpretation: The paper identifies a new policy margin: attracting and retaining relational buyers can raise supplier surplus in developing-country export sectors.

3 Data and measurement

3.1 Bangladeshi garment export data

- The analysis focuses on woven garments, especially trousers and shirts, exported by Bangladeshi manufacturers.
- Customs data provide detailed order-level information on buyers, sellers, product categories, quantities, prices, destinations, and timing.
- Buyers include major global retailers and brands such as H&M, Walmart, VF Corporation, The Gap, Levi Strauss, J.C. Penney, and others.
- Woven garments and knitwear are produced by largely different exporter sets. This separation lets the authors measure buyer sourcing strategies in excluded knitwear products and use them to explain woven-garment outcomes.

Interpretation: The product separation is central to identification. The buyer’s relationality measure is constructed outside the main outcome sample, reducing mechanical endogeneity.

3.2 Order-level input data

- The paper observes the type, price, and amount of fabric used to produce specific export orders.
- It also observes sewing-line labor measures for a subset of factories, including labor utilization, efficiency, and worker characteristics.
- Worker surveys and internal HR records provide information on worker wages, skills, gender, education, experience, and cognitive skills.

Interpretation: These data are unusual because they connect output orders to the inputs used for those same orders. This lets the paper test whether higher prices reflect higher costs or higher markups.

3.3 Buyer sourcing metric

The paper builds a buyer-level relational sourcing measure from supplier concentration. For buyer b , product j , and year t ,

$$\text{Relational}_{bjt} = -\frac{\#Sellers_{bjt}}{\#Shipments_{bjt}}. \quad (1)$$

The buyer-level metric is a volume-weighted average across product-year cells:

$$\text{Relational}_b = \sum_{jt \in J_b} \frac{Q_{bjt}}{Q_b} \text{Relational}_{bjt}. \quad (2)$$

In the main empirical analysis, the metric is computed on excluded products J^- and used to explain outcomes in included woven products J^+ :

$$\text{Relational}_b = \sum_{jt \in J^-} \frac{Q_{bjt}}{Q_{bJ^-}} \text{Relational}_{bjt}. \quad (3)$$

- The measure ranges from -1 to 0 .
- Values closer to 0 indicate more relational sourcing because the buyer uses fewer suppliers per shipment.
- Values closer to -1 indicate spot sourcing because the buyer spreads orders across more suppliers.

Interpretation: The key idea is that relational buyers concentrate orders with fewer suppliers, while spot buyers distribute orders across many potential suppliers.

3.4 Markup measurement

Let order o specify output quantity Q_o , output price P_o , fabric quantity F_o , and fabric price P_o^f . Let θ_o be the output elasticity with respect to fabric and λ_o be marginal cost. The fabric first-order condition implies

$$\lambda_o = \frac{P_o^f F_o}{\theta_o Q_o}, \quad M_o \equiv \frac{P_o}{\lambda_o} = \theta_o \frac{P_o Q_o}{P_o^f F_o}. \quad (4)$$

Define the observable term

$$\psi_o = \frac{P_o Q_o}{P_o^f F_o}. \quad (5)$$

For two orders o and o' , the log markup difference is

$$\ln M_o - \ln M_{o'} = \ln \psi_o - \ln \psi_{o'} + \ln \theta_o - \ln \theta_{o'}. \quad (6)$$

If fabric elasticity is common within seller-product-time cells, $\theta_o = \theta_{sjt}$, then seller-product-time fixed effects absorb the unobserved elasticity and $\ln \psi_o$ identifies within-cell markup differences.

Interpretation: This is the econometric bridge from order-level input data to markups. The paper does not need to know markup levels exactly; it identifies markup differences across buyers within tightly defined production cells.

4 Model and economic mechanism

4.1 Spot sourcing

- Under spot sourcing, buyers allocate orders through short-term competition.
- Suppliers compete for orders and are paid low prices.
- Spot sourcing works when production proceeds normally, but it gives suppliers weak incentives to take costly actions after unexpected capacity or delivery shocks.
- If a shock occurs, the supplier may fail to deliver or deprioritize the order.

Interpretation: Spot sourcing economizes on relationship investment but performs poorly when reliability requires noncontractible supplier effort.

4.2 Relational sourcing

- Relational buyers invest in organizational capabilities that let them make credible promises of future rents.
- They concentrate orders among fewer suppliers and sustain relationships over time.
- They pay higher prices and markups to suppliers, making it worthwhile for suppliers to exert effort or prioritize delivery when shocks occur.

Interpretation: Relational sourcing trades higher prices for reliable supply. The higher markup is not a mistake; it is the incentive payment that supports noncontractible performance.

4.3 Main prediction

The main testable prediction is:

$$\text{Relational buyer} \Rightarrow \text{higher supplier price and markup within seller-product-time cells.} \quad (7)$$

The model also predicts:

- relational buyers trade with fewer sellers than spot buyers;
- suppliers earn higher profits from relational buyers;
- suppliers prioritize relational buyers when supply disruptions occur;
- too few buyers may invest in relational sourcing because relational buyers create pecuniary externalities for suppliers.

Interpretation: The empirical analysis is designed to test the central price and markup prediction and then examine supporting mechanisms.

5 Empirical specifications

5.1 Price specification

The baseline price regression is conceptually

$$p_{sbjo} = \beta \text{Relational}_b + \delta_{sjt} + \delta_d + X'_{sbjo} \Gamma + \varepsilon_{sbjo}, \quad (8)$$

where

- p_{sbjo} is the log price of order o produced by seller s for buyer b in product j ;
- Relational_b is the standardized buyer relational sourcing metric;
- δ_{sjt} are seller-product-year fixed effects;
- δ_d are destination fixed effects;
- X_{sbjo} includes buyer, relationship, and order controls.

Interpretation: Seller-product-year fixed effects compare orders produced by the same exporter in the same product and year for different buyers.

5.2 Input-use specification

The same framework is estimated using fabric and labor outcomes:

$$\text{InputUse}_{sbjo} = \beta \text{Relational}_b + \delta_{sjt} + \delta_d + X'_{sbjo} \Gamma + \varepsilon_{sbjo}. \quad (9)$$

Outcomes include fabric price, buy-to-ship ratio, garment complexity, sewing line standard minute value, efficiency, number of workers, and labor share.

Interpretation: If relational buyers pay higher prices because their orders are more costly, Relational_b should predict higher input prices or higher input quantities. The paper finds little support for this.

5.3 Markup specification

Using the observed markup component ψ_o , the markup regression is

$$\mu_{sbjo} = \beta \text{Relational}_b + \delta_{sjt} + \delta_d + X'_{sbjo} \Gamma + \varepsilon_{sbjo}, \quad (10)$$

where μ_{sbjo} is the order-level log markup factor, net of the unobserved fabric elasticity through seller-product-time fixed effects.

Interpretation: This is the key test. The coefficient β measures whether suppliers earn higher markups on orders produced for relational buyers.

5.4 VF event-study specification

To complement cross-buyer evidence, the paper studies VF Corporation's shift toward a more relational sourcing strategy. The event-study specification is conceptually

$$y_{sbjo} = \sum_t \beta_t \mathbf{1}\{Year = t\} \times VF_b + \delta_{sjt} + X'_{sbjo} \Gamma + \varepsilon_{sbjo}. \quad (11)$$

Interpretation: The VF analysis follows a single buyer changing sourcing strategy. This addresses the concern that relational sourcing is only a fixed buyer characteristic.

6 Identification and empirical design

6.1 Why buyer-level measurement matters

- Buyer sourcing strategy is measured using transactions outside the main woven-garment sample.
- The metric is stable across products and origins, supporting the interpretation that it reflects buyer-level organizational capabilities.
- Buyer fixed effects explain a large share of variation in sourcing strategies across product-country pairs.

Interpretation: This supports treating sourcing strategy as a buyer-level attribute rather than a relationship-specific residual.

6.2 Within seller-product-time comparisons

- Seller-product-year fixed effects hold constant the supplier, product, and year.
- Destination fixed effects and buyer, relationship, and order controls address additional heterogeneity.
- Robustness checks use month, quarter, destination-year, seller-product-quarter, and other fixed effects.

Interpretation: The key identifying comparison is across buyers for highly similar orders produced by the same supplier.

6.3 Remaining concerns

- Relational buyers might order higher quality products not fully captured in HS codes or controls.
- Fabric elasticities might vary across buyer orders.
- Buyers might use sourcing intermediaries that create measurement error.
- Sourcing relationships might change over time, while the main buyer metric is time-invariant.

Interpretation: The paper addresses these concerns with input-use tests, product-complexity controls, robustness checks, alternative sourcing measures, and the VF event study.

7 Tables and figures

7.1 Table I and Figure I: buyer sourcing strategies

Read:

- Table I lists the largest 25 buyers of woven garments in Bangladesh and compares market shares, average number of sellers, relational ranking, and price ranking.
- H&M, The Gap, and Levi Strauss are near the top of the relational ranking, while several discount retailers rank lower.
- Figure I shows that buyer sourcing strategies are positively correlated across sourcing countries.
- This supports the view that relational sourcing is a buyer-level organizational capability rather than a purely country- or supplier-specific relationship.

Economic message: Buyers differ persistently in sourcing style. These differences are meaningful and align with industry descriptions.

TABLE I
BUYERS' CONCENTRATION AND SOURCING

	Market share %	Sellers per year average	Relational ranking	Price (residuals) ranking
Top 25 buyers	(1)	(2)	(3)	(4)
H&M Hennes and Mauritz	5.22	60.75	3	2
Wal Mart Stores	5.00	59.75	17	16
VF Corporation	4.14	25.50	5	17
The Gap Inc.	3.44	27.13	1	1
C & A Buying	3.17	43.38	8	9
K Mart Corporation	3.08	62.13	16	14
PVH Corporation	3.11	39.88	7	15
Levi Strauss & Co	2.21	7.50	2	7
J.C. Penney	1.96	25.88	11	10
Primark	1.42	23.00	10	24
Kik Textilen	1.32	51.88	25	22
Tesco	1.25	23.50	12	19
Kohls Department Stores Inc.	1.25	16.50	13	5
Asda	1.21	19.63	6	8
Marks & Spencer	1.15	10.25	4	11
Carrefour	1.13	26.88	14	18
G. Goldenpfennig GmbH	0.87	32.38	24	20
Tema Magazacilik	0.91	44.13	21	4
Public Clothing Company Inc.	0.84	25.25	23	23
Target Stores	0.85	19.50	15	12
Inditex (Zara)	0.81	33.13	20	3
Auchan S.A.	0.71	30.63	19	21
Charles Vogele	0.69	17.38	18	13
The Children's Place	0.68	11.13	9	6
IFG Corporation	0.65	14.38	22	25
Top 100 (market share = 66%)				
Mean	0.66	17.53		
Median	0.29	13.00		
Std. deviation	0.99	14.58		
Coeff. variation	1.49	0.83		
All buyers (N = 1,578)				
Mean	0.06	4.60		
Median	0.01	3.00		
Std. deviation	0.30	6.16		

Table I: buyer concentration and sourcing

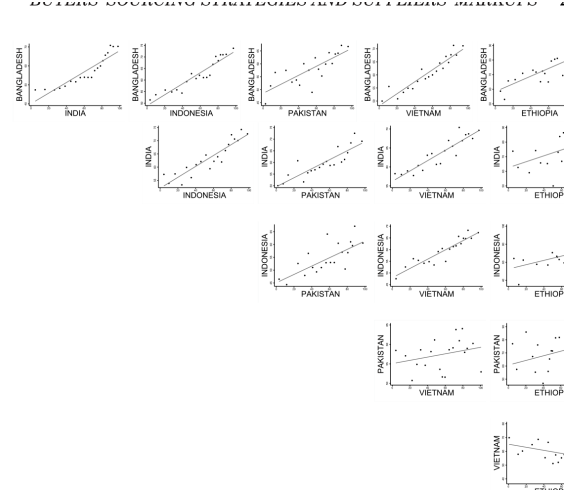


Figure I: sourcing strategies across countries

7.2 Table II and Table III: descriptive statistics and price results

Read:

- Table II reports summary statistics at the order, seller, buyer, and relationship levels.
- Orders are numerous and heterogeneous, and buyers/sellers have substantial variation in numbers of trading partners and relationship ages.

- Table III shows that a one standard deviation increase in relational sourcing is associated with about 2 log points higher order prices.
- The effect survives seller-product-year fixed effects, destination controls, buyer controls, relationship controls, and order controls.

Economic message: The descriptive data confirm the richness of buyer-seller relationships, and Table III establishes the first main result: relational buyers pay higher prices.

TABLE II
SUMMARY STATISTICS

	Obs.	Mean	Std. dev.	P10	P25	P50	P75	P90
Panel A: Orders								
<i>Buy-to-Ship_o</i>	22,741	0.87	0.29	0.51	0.67	0.86	1.04	1.22
<i>Length_o</i> (months)	22,741	4.24	3.25	1.47	2.17	3.3	5.23	8.03
Panel B: Sellers								
<i>Count_{st}^o</i>	3,165	14.60	13.07	3	6	11	19	29
<i>Count_{sjt}^o</i>	6,872	6.03	7.53	1	2	3	7	14
<i>Count_{st}^t</i>	3,165	3.27	1.88	1	2	3	4	6
<i>Share_{st}^e</i>	3,165	57.76	34.92	6.40	24.29	62.30	92.67	100
<i>Count_{st}^b</i>	500	20.97	17.10	4	8.5	17	28	42.5
<i>Count_{st}^b</i>	3,165	5.88	4.86	1	2	5	8	12
<i>Count_{sjt}^b</i>	6,872	2.91	2.91	1	1	2	4	6
<i>Share_{st}^e</i>	3,165	43.98	36.91	1.71	8.50	34.67	82.18	100
<i>Length_s</i> (years)	500	6.65	1.54	4.08	5.75	7.63	7.75	7.75
Panel C: Buyers								
<i>Count_{bt}^o</i>	4,478	13.37	29.75	1	2	5	13	27
<i>Count_{bjt}^o</i>	8,070	5.75	11.54	1	1	2	5	12
<i>Count_{bt}^t</i>	4,478	4.24	3.83	1	2	3	5	9
<i>Share_{bt}^e</i>	4,478	59.47	35.71	5.96	26.41	63.58	100	100
<i>Count_{bt}^b</i>	2,529	54.40	50.06	9	18	37	72	137
<i>Count_{bt}^t</i>	7,569	22.05	20.52	4	7	14	30	58
<i>Count_{bjt}^t</i>	11,942	8.80	9.07	1	3	5	12	21
<i>Share_{bt}^e</i>	4,478	48.62	37.97	0	11.72	42.08	92.28	100
<i>Length_b</i> (years)	1,578	5.48	2.42	1.58	3.58	6.42	7.67	7.75
Panel D: Relationships								
<i>Count_{sb}^o</i>	10,448	3.38	4.58	1	1	2	4	7
<i>Count_{sbjt}^o</i>	12,858	2.52	3.14	1	1	1	3	5
<i>Count_{sb}^t</i>	10,448	1.46	0.85	1	1	1	2	2
<i>Length_{sb}</i> (years)	5,658	1.87	2.03	0.08	0.25	1.17	2.75	5.08

Notes. Super- and subscripts are as follows: *o* corresponds to orders, *b* to buyers, *s* to sellers, *j* to HS6 product categories, and *t* to years. *Count_{st}^o* is the number of *x* per *y*. For example, *Count_{st}^o* is the number of orders per seller-product-year combination. *Length_o* is the number of months between the first import shipment and the last export shipment of the order. *Length_{sb}*, *Length_{bt}*, and *Length_s* are the number of years the buyer-seller pair, buyer, and seller are observed trading in the dataset, respectively. A value of 7.75 in these variables

Table II: summary statistics

TABLE III
BUYERS' SOURCING AND PRICES

	<i>P_{sbjo}</i>			
	(1)	(2)	(3)	(4)
<i>Relational_b</i>	0.020*** (0.008)	0.023*** (0.008)	0.019** (0.009)	0.021*** (0.008)
FEs	<i>sjt</i>	<i>sjt, d</i>	<i>sjt, d</i>	<i>sjt, d</i>
Controls	.	B	B,R	B,R,O
<i>R</i> ²	0.57	0.59	0.63	0.73
Obs.	18,664	18,513	15,647	15,647

Notes. Standard errors are in parentheses, clustered at the buyer level. The outcome in all regressions is the log price of an order between a seller and a buyer in a given product category, *P_{sbjo}*. The main regressor in all cases is the baseline, buyer-specific metric of relational sourcing and it is standardized. Column (1) includes our baseline fixed effect, defined at the level of the seller-product-year triplet. Columns (2)–(4) sequentially add buyer-, relationship-, and order-level covariates, as follows. Buyer controls (B): fixed effect for the main destination of the buyer, cohort of the buyer (year first observed in the data), size (log volume imported by the buyer throughout our data and across all woven products), age of the buyer at the time of the order (log number of months elapsed since first observed in the data), and a dummy indicating whether the buyer is a signatory of the accord as of 2019. Relationship controls (R): Cohort of the relationship (year first observed in the data), size (log volume traded by the buyer and seller throughout our data and across all woven products), age of the relationship at the time of the order (log number of months elapsed since first observed in the data), share of the seller in all of the buyer's trade, and share of the buyer in all of the seller's trade. Order controls

Table III: relational sourcing and prices

7.3 Figure II and Table IV: price robustness and input use

Read:

- Figure II reports 522 robustness estimates of the relational sourcing coefficient in price regressions.
- Almost all specifications yield positive coefficients, and a majority are statistically significant.
- Table IV shows that relational sourcing does not systematically predict fabric price, buy-to-ship ratio, or product complexity once controls are included.

Economic message: Higher prices paid by relational buyers are not explained by obvious fabric or complexity differences.

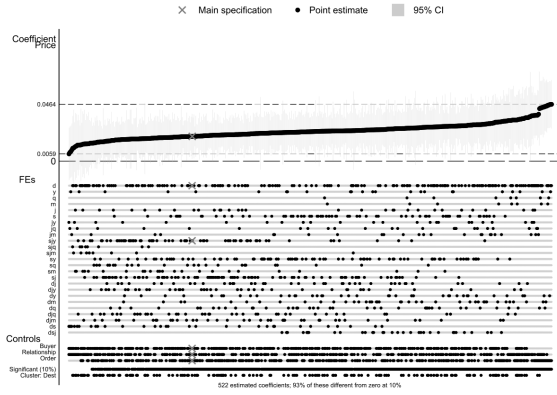


FIGURE II

Robustness of Price Result to Alternative Fixed Effects and Controls

The graph presents 522 estimates of the coefficient on the buyer-specific variable. Figure II: price robustness

TABLE IV
BUYERS' SOURCING AND INPUT USAGE

	p'_{sbjo}		$\left(\frac{F}{Q}\right)_{sbjo}$		$Complex_{sbjo}$	
	(1)	(2)	(3)	(4)	(5)	(6)
$Relational_b$	0.008 (0.006)	0.003 (0.007)	-0.004 (0.006)	-0.007 (0.007)	0.021* (0.012)	-0.003 (0.011)
FES	sjt	sjt, d	sjt	sjt, d	sjt	sjt, d
Controls	.	B,R,O	.	B,R,O	.	B,R,O
R^2	0.64	0.69	0.36	0.44	0.46	0.58
Obs.	18,664	15,647	18,664	15,647	18,664	15,647

Notes. Standard errors are in parentheses, clustered at the buyer level. The main regressor in all cases is the baseline, buyer-specific metric of relational sourcing and it is standardized. Outcomes are: the log weighted average price of fabric in the order, p'_{sbjo} (columns (1) and (2)), the buy-to-ship ratio of the order, $\left(\frac{F}{Q}\right)_{sbjo}$ (columns (3) and (4)), and a measure of complexity of the garment order (the log of the number of fabric types used for producing the order), $Complex_{sbjo}$ (columns (5) and (6)). All columns feature seller-product-year fixed effects. In addition, even-numbered columns also include buyer-, relationship-, and order-level controls, as follows. Buyer controls (B): fixed effect for the main destination of the buyer, cohort of the buyer (year first observed in the data), size (log volume imported by the buyer throughout our data and across all woven products), age of the buyer at the time of the order (log number of months elapsed since first observed in the data), and a dummy indicating whether the buyer is a signatory of the accord as of 2019. Relationship controls (R): Cohort of the relationship (year first observed in the data), size (log volume traded by the buyer and seller throughout our data and across all woven products), age of the relationship at the time of the order (log number of months elapsed since first observed in the data), share of the seller in all

Table IV: fabric and order input usage

7.4 Table V and Figure III: labor inputs and markup robustness

Read:

- Table V studies sewing-line labor outcomes such as SMV, efficiency, number of workers, and labor shares.
- Relational sourcing does not imply higher sewing labor costs, more skilled workers, or lower efficiency.
- Figure III reports 522 robustness estimates for markup regressions.
- Markup effects are positive across virtually all specifications, with the baseline effect around 0.026 log points.

Economic message: The labor evidence further rules out cost explanations, while Figure III shows the markup result is robust to many fixed-effect and control combinations.

BUYERS' SOURCING STRATEGIES AND SUPPLIERS' MARKUPS 24

	SMV_{dbs}		$Efficiency_{dbs}$		$\#Workers_{dbs}$		$Share\ Helpers_{dbs}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Relational_b$	-0.084 (0.353)	-0.024 (0.320)	-0.010* (0.006)	-0.005 (0.005)	0.428 (0.534)	0.396 (0.335)	0.001 (0.002)
FES	$smv(\tau), \tau$	$smv(\tau), d, \tau$	$smv(\tau), \tau$	$smv(\tau), d, \tau$	$smv(\tau), \tau$	$smv(\tau), d, \tau$	$smv(\tau), d, \tau$
R^2	0.77	0.85	0.20	0.23	0.80	0.92	0.94
Obs.	155,723	155,723	116,905	116,896	125,940	125,932	125,932

Notes. Standard errors are in parentheses, clustered at the level of the buyer and production line. Across all specifications, the regressor of interest is the metric on relational sourcing, standardized and increasing in the relational characteristic of the buyer. The outcome in columns (1) and (2) is the standard minutes value (SMV), defined as the amount of time a particular garment is supposed to take to be sewed together compared by the factory's industrial engineers (also based on international literature of SMV of elemental sewing processes). Columns (3) and (4) study labor efficiency of a particular line in a plant, producing for a buyer on a given day. $Efficiency_{dbs}$: Labor efficiency is constructed as the ratio between the minutes equivalent of the output and the minutes of labor input. In turn, the output is calculated as standard minute values times the number of pieces and the input is calculated using the number of workers times the run time. See the main text for a comprehensive description. The outcome in columns (5) and (6) is the number of workers active on the line. $\#Workers_{dbs}$, and in columns (7) and (8) it is the share of each workers that are line helpers. $Share\ Helpers_{dbs}$. The discrepancies in sample size across columns are due to the fact that not all plants keep administrative records of all labor usage metrics studied here. All specifications include as controls for relevant buyer characteristics (in size as a garment importer in Bangladesh, whether the buyer is a signatory of the compliance accord as of 2019, and the cohort of the buyer). Odd-numbered columns condition on fixed effects corresponding to the seller-month (and τ) and the day (τ). Even-numbered columns, in addition, include a fixed effect for the production line of the seller (d). * $p < .10$, ** $p < .05$, *** $p < .01$.

Table V: sourcing and labor usage

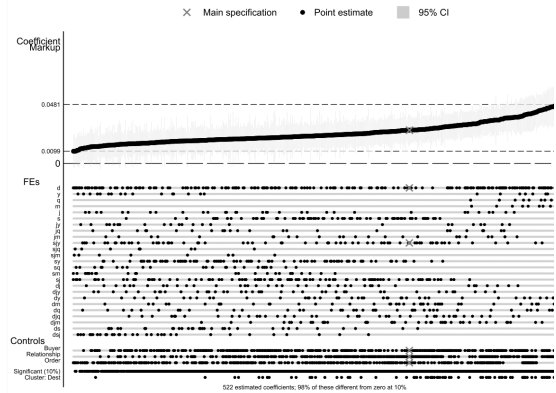


FIGURE III

Robustness of Markups Result to Alternative Fixed Effects and Controls

The graph presents 522 estimates of the coefficient on the buyer-specific variable. Figure III: markup robustness

7.5 Table VI and Figure IV: markup decomposition and VF event study

Read:

- Table VI decomposes the price effect into marginal cost and markup components.

- Relational sourcing raises prices, does not raise estimated marginal costs, and raises markup factors.
- The baseline markup effect is about 0.026 log points and remains robust to controls for seasonality, product specialization, quality, and dropping largest buyers or sellers.
- Figure IV studies VF Corporation’s change in sourcing strategy.
- After VF shifts toward relational sourcing around 2010, suppliers earn higher markups on orders for VF relative to comparison buyers.

Economic message: The core mechanism is confirmed: relational sourcing increases supplier markups. The VF evidence supports the interpretation that changing sourcing strategy changes markups.

TABLE VI
BUYERS’ SOURCING, MARKUPS, AND COSTS

	$P_{d,p}$ (1)	$MC_{d,p}$ (2)	$\mu_{d,p}$ (3)	$\mu_{d,p}$ (4)	$\mu_{d,p}$ (5)	$\mu_{d,p}$ (6)	$\mu_{d,p}$ (7)	$\mu_{d,p}$ (8)
Relationships	0.021*** (0.008)	-0.005 (0.007)	0.026*** (0.007)	0.026*** (0.007)	0.026*** (0.007)	0.032*** (0.010)	0.039*** (0.010)	0.027*** (0.012)
FFs	gft, d	gft, d	gft, d	gft, d	gft, d	gft, d, q	gft, d	gft, d
Controls	B.R.O	B.R.O	B.R.O	B.R.O	B.R.O	B.R.O	B.R.O	B.R.O
Robustness				Season	Product	Quality	Small b	Small s
R^2	0.73	0.62	0.41	0.41	0.41	0.59	0.43	0.44
Obs.	15,647	15,647	15,647	15,647	15,647	10,103	10,780	7,479

Notes. Standard errors are in parentheses, clustered at the buyer level. The outcome in column (1) is the log price of an order between a seller and a buyer in a given product category, $p_{d,p}$. The outcome in column (2) is the estimated log marginal cost of the order, $mc_{d,p}$. In all other columns, the outcome is the log markup factor, $\mu_{d,p}$. The main regressor in all cases is the baseline, buyer-specific metric of relational sourcing and it is standardized. All columns but (6) include seller-product-year and destination fixed effects. As such, column (1) simply reproduces the results of column (4) in Table VI and columns (2) and (3) use the same specification to study marginal costs and markups. All remaining columns report the results of different robustness exercises, for brevity shown only on $\mu_{d,p}$. All columns in the table include buyer (B), relationship (R), and order-level (O) covariates, as follows. Buyer controls (B): fixed effect for the main destination of the buyer, cohort of the buyer (year first observed in the data), size (log volume imported by the buyer throughout our data and across all seven products), age of the buyer at the time of the order (log number of months elapsed since first observed in the data), and a dummy indicating whether the buyer is a subsidiary of the parent as of 2010. Relationship controls (R): Cohort of the relationship (year first observed in the data), size (log volume traded by the buyer and seller throughout our data and across all seven products), age of the relationship at the time of the order (log number of months elapsed since first observed in the data), share of the order in all of the buyer’s trade, and share of the buyer in all of the seller’s trade. Order controls (O): Cohort of the order (log volume) and log price of fabric of the order. Column (4) controls for seasonality patterns, product specialization, and the physical quality of the order measured as follows. Season: Herfindahl index describing how concentrated the trade in a relationship is in one season, the share of the largest season in the seller-buyer-year combination, and an indicator that picks up orders in each season. Product: defined analogously to controls described for seasonality. Quality: measure of complexity of the garment order (the log of the number of fabric types used for producing the order, elsewhere labeled as Complexity), and a fixed effect for the seller-product-year (fabric-type) pair, exploiting the type and origin of fabric to define the variety of the order. A category is as specific as Nike Ltd.’s men’s shirts made of w.o. fab. containing > 85% cotton, printed, plain weaves, weighing more than 100 g/m² but not more than 200 g/m² sourced from India in 2010. Column (7) trims the sample to drop all the orders of the largest buyer of the seller-year-product. Column (8), marginally, drops all orders of the largest seller of the buyer in the product-year combination. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table VI: prices, marginal costs, and markups

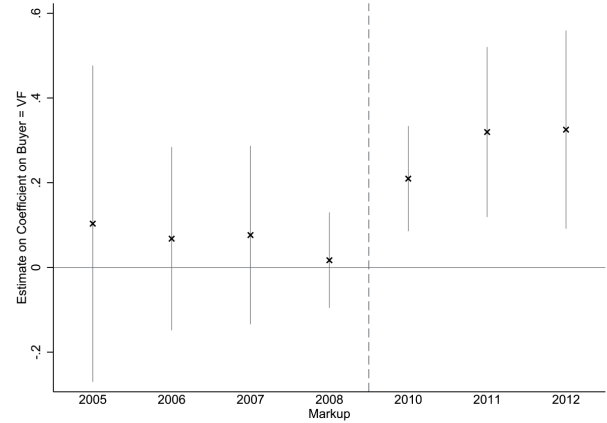


FIGURE IV
Figure IV: VF event study

8 Result synthesis

8.1 Buyer sourcing as organizational capability

The sourcing metric is correlated across countries and products. This suggests that sourcing style is embedded in buyer organization: procurement routines, relationship management, tolerance for concentration risk, internal coordination, and ability to make credible promises. The paper’s buyer-level interpretation is therefore empirically plausible.

8.2 Price versus cost decomposition

Relational buyers pay higher prices. The paper then asks whether these prices reflect higher costs. Fabric use, fabric prices, fabric efficiency, garment complexity, sewing-line efficiency, and worker characteristics do not increase for relational buyers. The remaining explanation consistent with the production framework is a higher markup.

8.3 Relational sourcing and supplier rents

The model predicts that suppliers receive rents from relational buyers because those rents incentivize costly noncontractible actions. The empirical estimates align closely with this prediction: higher prices translate into higher markups, not higher costs.

8.4 Policy interpretation

If relational buyers help suppliers earn rents and may induce reliability, upgrading, or stability, their presence can have broader benefits. Since buyer relational capabilities are partly organizational, development policy

might target buyer attraction and relationship formation, not only supplier capability building.

9 Short summary

- The paper studies Bangladeshi garment exporters and their global buyers.
- Buyers differ between relational sourcing and spot sourcing.
- Relational sourcing is measured as concentration of a buyer's orders among fewer suppliers.
- The sourcing metric is stable across products and countries, suggesting buyer-level capabilities.
- Relational buyers pay higher order prices for comparable products from the same suppliers.
- The paper uses novel order-level fabric and sewing labor data to test whether higher prices reflect higher variable costs.
- Fabric and labor costs do not explain the higher prices.
- A production-based markup framework shows that relational buyers pay higher markups.
- The markup results are robust across many fixed-effect structures, controls, and samples.
- A VF Corporation event study supports the interpretation that shifts toward relational sourcing increase markups.
- The paper contributes to global value chain, markup, relational contracting, and development policy literatures.

10 Conclusion

10.1 Core conclusion

Cajal-Grossi, Macchiavello, and Noguera show that relational buyers in global garment supply chains pay higher markups to suppliers than spot buyers. These markups are not compensation for more expensive fabric, more labor, or more complex production. Instead, they are consistent with relational incentives that support reliable supply.

10.2 Economic intuition

Spot buyers use competitive pressure to minimize prices. Relational buyers pay higher markups to sustain long-term relationships and induce suppliers to deliver reliably even when shocks occur. The supplier rents are therefore part of an incentive system, not merely a bargaining anomaly.

10.3 Takeaway

Buyer identity matters for development and trade. Exporters do not merely benefit from access to foreign demand; they benefit from relationships with buyers whose sourcing capabilities allow them to sustain relational contracts. Attracting relational buyers may raise supplier surplus and improve the functioning of global value chains.